

$$= \text{constant} \times t$$

8.

Ans: D

Using COE:

$$E_i = E_f$$

$$GPE_i = KE_f$$

$$mgr = \frac{1}{2}mv^2$$

$$2mgr = mv^2 \quad \text{----- (1)}$$

$$T_{\text{bottom}} - mg = \frac{mv^2}{r}$$

$$T_{\text{bottom}} = \frac{mv^2}{r} + mg \quad \text{sub (1) into eqn}$$

$$= \frac{2mgr}{r} + mg$$

$$= 3mg$$

8

Ans: D